

Bike Stations Sharing Data Analysis

1. Project Overview and Objective

Project Overview

The Bike Sharing Station Analytics Dashboard is an interactive Power BI project designed to analyze the operational performance of a public bike-sharing network. The dashboard provides insights into station availability, bike utilization, contract performance, and geographic distribution of bike stations. By transforming raw station data into meaningful visualizations and KPIs, the project helps monitor network efficiency, identify underperforming stations, and support data-driven operational decisions.

Project Objective

The primary objective of this project is to analyze bike-sharing station data to evaluate operational performance and improve resource utilization. The dashboard focuses on monitoring station availability, bike usage, contract-wise performance, and geographic coverage to help stakeholders optimize bike allocation, enhance customer service, and improve the overall efficiency of the bike-sharing system.

Key Objectives

- Analyze the operational status of bike-sharing stations.
 - Monitor bike availability and station capacity utilization.
 - Compare contract-wise station performance.
 - Identify stations with the highest and lowest bike availability.
 - Analyze the geographic distribution of bike stations.
 - Track key operational KPIs such as available bikes, bikes in use, and available bike stands.
 - Support data-driven decision-making for better resource planning and network optimization.
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2. Data Sources:

Source Description and Timeline:

- **Source:** Open Government Data (Bike Sharing System Data)
- **Description:** The dataset contains information about public bike-sharing stations, including station name, geographic location, operational status, available bikes, available bike stands, total bike stands, and last update time. It is used to analyze station utilization, operational efficiency, and bike availability.
- **Timeline:** 2020–2025 (Sample/Recent Open Data)

Domain

- Smart Transportation / Urban Mobility / Public Bike Sharing
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3. Problem Statement:

- **Analyze** station-wise bike availability to identify stations with consistently high and low bike availability.
- **Examine** the distribution of bike stations across cities to identify locations with the largest station networks.
- **Measure** the operational status of bike stations by calculating the percentage of stations currently in service.
- **Identify** stations requiring urgent bike redistribution by detecting stations with no available bikes.
- **Evaluate** stations that have reached full capacity to determine where bike collection is required.

- **Assess** station capacity and utilization to identify high-capacity stations and evaluate how effectively their capacity is being used.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Number	Integer	Unique identification numbers are assigned to each bike station.
Name	Text	Name of the bike-sharing station.
Address	Text	Physical address of the bike station.
Position	Geographic (Latitude & Longitude)	Geographic coordinates used to identify and map the station location.
Contract Name	Text	Name of the city or service area where the station operates.
Status	Text	Operational status of the station (e.g., OPEN or CLOSED).
Bike Stands	Integer	Total number of bike docking stands available at the station (station capacity).
Available Bikes	Integer	Number of bikes is currently available for users to rent.
Available Bike Stands	Integer	Number of empty dockings stands available for returning bikes.
Banking	Boolean (True/False)	Indicates whether the station supports banking/payment services.
Bonus	Boolean (True/False)	Indicates whether the station is designated as a bonus station for incentive programs.
Last Update	Date/Time	Date and time when the station information was last updated.

5. Tools & Technologies

- **Power BI:** Performed data modeling, created DAX measures and calculated columns, designed interactive dashboards, developed KPI cards, charts, maps, slicers, and visualizations to analyze bike station availability, operational status, capacity utilization, and city-wise performance.

6. Data Pre-Processing:

Tasks Performed:

- **Data Cleaning & Transformation:** Cleaned the dataset by removing duplicates, handling missing values, standardizing data formats, and creating calculated columns to improve data quality and support accurate analysis.
- **Filtering & Sorting:** Applied filtering and sorting techniques to organize data, focus on relevant records, and streamline the analytical process.

7. Data Cleaning and DAX:

Calculated Columns & Measures: Created DAX calculated columns and measures to derive KPIs such as Operational Rate, Capacity Utilization %, Available Bikes, Empty Stations, and Full Stations.

Performance Optimization: Optimized the data model by removing unnecessary columns, using appropriate data types, and organizing tables into a star schema where applicable to improve report performance.

SCREENSHOT OF DATA CLEANING:

The Data Cleaning Process in Power Query

- **Import and Access Data:** Start by importing your data into Power BI Desktop using Get Data. Then, click Transform Data to open the Power Query Editor, where all cleaning will take place.
 1. **Cleaning, Trim and Standardize Text:** Use Transform > Format to change text to lowercase/upercase and used Clean and Trim to remove extra space.

The screenshot shows the Power Query Editor interface. On the left, the 'Query Settings' pane is open, displaying the 'APPLIED STEPS' list. The steps are: Source, Navigation, Promoted headers, Changed column type, Renamed Columns - Number, Cleaned Text, Trimmed Text, and 'Changed Type' (which is selected and highlighted with a blue bar). On the right, a preview of the data is shown. The column 'Number' is selected, and its data type is 'Number'. The preview shows a list of 10 rows with values ranging from 1 to 10, and corresponding numerical values in the adjacent column.

Index	Number
1	2000
2	1297
3	8265
4	8065
5	1011
6	4342
7	6176
8	3322
9	1264
10	1191

2.

The screenshot shows the Power Query Editor interface. On the left, the 'Query Settings' pane is open, displaying the 'APPLIED STEPS' list. The steps are: Source, Navigation, Promoted headers, Changed column type, Renamed Columns - Number, Cleaned Text, Trimmed Text, 'Changed Type' (which is selected and highlighted with a blue bar), Renamed Columns - Name, Cleaned Text1, Trimmed Text1, and 'Changed Type1'. On the right, a preview of the data is shown. The column 'Name' is selected, and its data type is 'Text'. The preview shows a list of 10 rows with values ranging from 1 to 10, and corresponding text values in the adjacent column.

Index	Name
1	2000 - TEST DSI PLAISIR
2	1297- ST FÉRRÉOL DAVSO
3	8265- NÉGRESKO - PAULET
4	8065-PROMENADE POMPIDOU PALM ...
5	01011 - BORNE TEST NANTES 1
6	4342 - FLAMMARION MONTRICHET
7	6176-CANTINI ROUET
8	3322 - ARENC CHANTERAC
9	1264 - CANEBIÈRE DUGOMMIER
10	1191-CANEBIERE-ST FERREOL

3. Address Missing and Duplicate Data:

- **Handle Missing Values:** Review column quality (View > Column quality). Decide to remove rows with missing values in key columns by filtering out `null` replace nulls with a default value (Unknown) (Transform > Replace Values)

Replace Values

Replace one value with another in the selected columns.

Value To Find

null

Replace With

Unknown

Advanced options

OK

Cancel

Query Settings

PROPERTIES

Name

Feuil1

All Properties

APPLIED STEPS

Source

Navigation

Promoted headers

Changed column type

Renamed Columns - Number

Cleaned Text

Trimmed Text

Changed Type

Renamed Columns - Name

Cleaned Text1

Trimmed Text1

Changed Type1

Renamed Columns - Address

Cleaned Text2

Trimmed Text2

Changed Type2

X Replaced Value

A ^B C Address	A ^B C Address
Valid	Valid 100%
Error	Error 0%
Empty	Empty 0%
	Unknown
ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL	ST FERREOL DAVSO - RUE FRANCIS DAVSO ANGLE RUE SAINT FERREOL
NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO
PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES PO...
Borne test 1	Borne test 1
FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...	FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMAR...
CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI	CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI
32 rue de Chanterac 13003 Marseille	32 rue de Chanterac 13003 Marseille
CANEBIERE DUGOMMIER - LA CANEBIERE ANGLE BOULEVARD DUGO...	CANEBIERE DUGOMMIER - LA CANEBIERE ANGLE BOULEVARD DUGO...
CANEBIERE SAINT FERREOL - ANGLE CANEBIERE SAINT FERREOL	CANEBIERE SAINT FERREOL - ANGLE CANEBIERE SAINT FERREOL
SAKAKINI SAINT PIERRE - BOULEVARD SAKAKINI ANGLE RUE SAINT PIE...	SAKAKINI SAINT PIERRE - BOULEVARD SAKAKINI ANGLE RUE SAINT PIE...
BLANCARDE - GARE DE LA BLANCARDE	BLANCARDE - GARE DE LA BLANCARDE
GAMBETTA LAFAYETTE - 6 ALLEES LEON GAMBETTA	GAMBETTA LAFAYETTE - 6 ALLEES LEON GAMBETTA

4.

ABC

Position

Valid

Error

Empty

100%

0%

0%

46.5488766782771, 15.6350265324683
43.293485736186526, 5.378423799723321
43.269318556371154, 5.389387542314263
43.267272515671856, 5.372088524255505
47.19504, -1.5575
43.305993453766845, 5.393678259953781
43.284775983612235, 5.385593763617925
43.3123370339008, 5.3687418043728
43.29789631329341, 5.381132215559973
43.29620767061392, 5.37693066220596
43.292800863545835, 5.401285286456447
43.29626340517055, 5.406000803377506
43.29902793826475, 5.381500046726694
43.28946760889984, 5.36370990786361
43.290329770206334, 5.375548341850035
43.2961387765145, 5.3707713851323
43.29582518327951, 5.385996090720193

Query Settings

PROPERTIES

Name

Feuil1

All Properties

APPLIED STEPS

Changed column type

Renamed Columns - Num...

Cleaned Text

Trimmed Text

Changed Type

Renamed Columns - Name

Cleaned Text1

Trimmed Text1

Changed Type1

Renamed Columns - Addr...

Cleaned Text2

Trimmed Text2

Changed Type2

Replaced Value

Renamed Columns - Positi...

Cleaned Text3

Trimmed Text3

Changed Type3

5.

Banking	
Valid	100%
Error	0%
Empty	0%
	FALSE
	TRUE
	TRUE
	TRUE
	FALSE
	TRUE
	TRUE
	TRUE
	TRUE
	TRUE

Query Settings

PROPERTIES

Name

Feuil1

All Properties

APPLIED STEPS

Trimmed Text2

Changed Type2

Replaced Value

Renamed Columns3 - Position

Cleaned Text3

Trimmed Text3

Changed Type3

Renamed Columns - Banking

Cleaned Text4

Trimmed Text4

Changed Type4

6.

Bonus	
Valid	100%
Error	0%
Empty	0%
	FALSE
	FALSE
	FALSE
	FALSE
	FALSE
	FALSE
	FALSE
	FALSE
	FALSE
	FALSE

Query Settings

PROPERTIES

Name

Feuil1

All Properties

APPLIED STEPS

Trimmed Text2

Changed Type2

Replaced Value

Renamed Columns3 - Position

Cleaned Text3

Trimmed Text3

Changed Type3

Renamed Columns - Banking

Cleaned Text4

Trimmed Text4

Changed Type4

Renamed Columns - Bonus

Cleaned Text5

Trimmed Text5

Changed Type5

7.

Status	
Valid	100%
Error	0%
Empty	0%
	OPEN
	OPEN
	OPEN
	OPEN
	OPEN
	OPEN
	OPEN
	OPEN
	OPEN
	OPEN
	OPEN
	OPEN

Query Settings

PROPERTIES

Name

Feuil1

All Properties

APPLIED STEPS

Trimmed Text2

Changed Type2

Replaced Value

Renamed Columns3 - Position

Cleaned Text3

Trimmed Text3

Changed Type3

Renamed Columns - Banking

Cleaned Text4

Trimmed Text4

Changed Type4

Renamed Columns - Bonus

Cleaned Text5

Trimmed Text5

Changed Type5

Renamed Columns - Status

Cleaned Text6

Trimmed Text6

Changed Type6

8.

A ^B C Contract Name		
Valid	100%	
Error	0%	
Empty	0%	
maribor		
marseille		
marseille		
marseille		
nantes		
marseille		
marseille		
marseille		
marseille		
marseille		
marseille		

A ^B C Contract Name		
Valid	100%	
Error	0%	
Empty	0%	
Maribor		
Marseille		
Marseille		
Marseille		
Nantes		
Marseille		
Marseille		
Marseille		
Marseille		
Marseille		

Query Settings

PROPERTIES

Name

Feuil1

All Properties

APPLIED STEPS

Cleaned Text3

Trimmed Text3

Changed Type3

Renamed Columns - Banking

Cleaned Text4

Trimmed Text4

Changed Type4

Renamed Columns - Bonus

Cleaned Text5

Trimmed Text5

Changed Type5

Renamed Columns - Status

Cleaned Text6

Trimmed Text6

Changed Type6

Cleaned Text7

Trimmed Text7

Capitalized Each Word

Changed Type7

9.

1 ² 3 Bike Stands		
Valid	100%	
Error	0%	
Empty	0%	
	1	
	9	
	15	
	15	
	0	
	10	
	24	
	20	
	10	
	20	

Query Settings

PROPERTIES

Name

Feuil1

All Properties

APPLIED STEPS

Changed Type3

Renamed Columns - Banking

Cleaned Text4

Trimmed Text4

Changed Type4

Renamed Columns - Bonus

Cleaned Text5

Trimmed Text5

Changed Type5

Renamed Columns - Status

Cleaned Text6

Trimmed Text6

Changed Type6

Cleaned Text7

Trimmed Text7

Capitalized Each Word

Changed Type7

Cleaned Text8 - Bike Stands

Trimmed Text8

Changed Type8

10.

1 ² 3 Available Bike Stands		
Valid	100%	
Error	0%	
Empty	0%	
	0	
	0	
	0	
	15	
	0	
	0	
	24	
	13	
	0	
	2	
	10	

Query Settings

PROPERTIES

Name

Feuil1

All Properties

APPLIED STEPS

Changed Type4

Renamed Columns - Bonus

Cleaned Text5

Trimmed Text5

Changed Type5

Renamed Columns - Status

Cleaned Text6

Trimmed Text6

Changed Type6 - Contact Name

Cleaned Text7

Trimmed Text7

Capitalized Each Word

Changed Type7

Cleaned Text8 - Bike Stands

Trimmed Text8

Changed Type8

Cleaned Text8 - Available Bike ...

Trimmed Text9

Changed Type9

11.

123 Available Bikes	
Valid	100%
Error	0%
Empty	0%
	0
	0
	0
	0
	0
	0
	0
	7
	0
	0
	0

12.

Last Update	
Valid	100%
Error	0%
Empty	0%
	02-Jun-22 3:14:38 PM +05:30
	06-Apr-22 12:54:35 PM +05:30
	27-Sep-22 10:57:48 AM +05:30
	08-Jun-22 11:50:13 AM +05:30
	23-May-22 8:11:50 PM +05:30
	13-Sep-22 11:28:01 AM +05:30
	06-Sep-22 6:02:17 PM +05:30
	14-Sep-22 1:27:09 PM +05:30
	12-Sep-22 11:18:28 AM +05:30
	03-Oct-22 11:34:01 AM +05:30
	13-Sep-22 3:41:24 PM +05:30

- **DAX Measures:** Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals.

1. Average Available Bike Stands = `AVERAGE('Feuil1'[Available Bike Stands])`
2. Average Available Bikes = `AVERAGE('Feuil1'[Available Bikes])`
3. Average Capacity = `AVERAGE('Feuil1'[Bike Stands])`
4. Banking Stations = `COUNTROWS(FILTER('Feuil1','Feuil1'[Banking] = TRUE()))`
5. Bikes In Use = `SUM('Feuil1'[Bike Stands]) - SUM('Feuil1'[Available Bikes])`
6. Bonus Stations = `COUNTROWS(FILTER('Feuil1','Feuil1'[Bonus] = TRUE()))`
7. Capacity Utilization % = `DIVIDE([Bikes In Use],[Total Bike Stands],0)`
8. Closed Station % = `DIVIDE([Closed Stations],[Total Bike Stands])`
9. Closed Stations = `COUNTROWS(FILTER('Feuil1','Feuil1'[Status] = "CLOSED"))`
10. Empty Stations = `COUNTROWS(FILTER('Feuil1','Feuil1'[Available Bikes] = 0))`
11. Full Stations = `COUNTROWS(FILTER('Feuil1','Feuil1'[Available Bike Stands] = 0))`
12. Open Station % = `DIVIDE([Open Stations],[Total Bike Stands])`
13. Open Stations = `CALCULATE(COUNTROWS('Feuil1'),'Feuil1'[Status] = "OPEN"))`
14. Open Stations 2 = `COUNTROWS(FILTER('Feuil1','Feuil1'[Status] = "OPEN"))`
15. Operational Rate = `DIVIDE([Open Stations],COUNTROWS(Feuil1))`

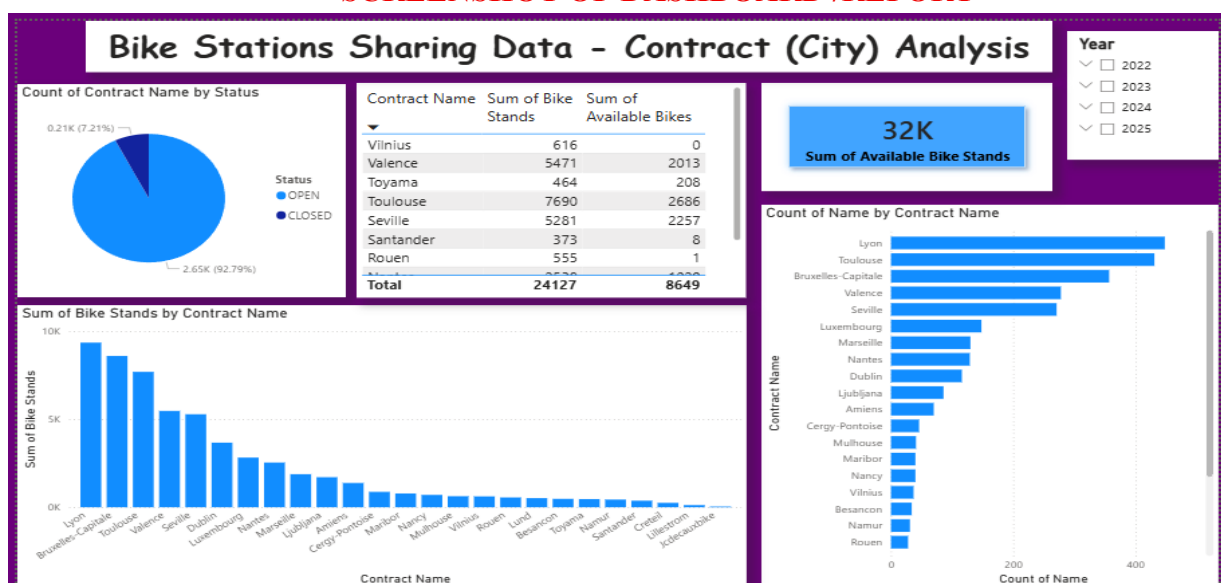
16. Total Available Bikes = `SUM('Feuill1'[Available Bikes])`
 17. Total Bike Stands = `COUNT(Feuill1[Bike Stands])`

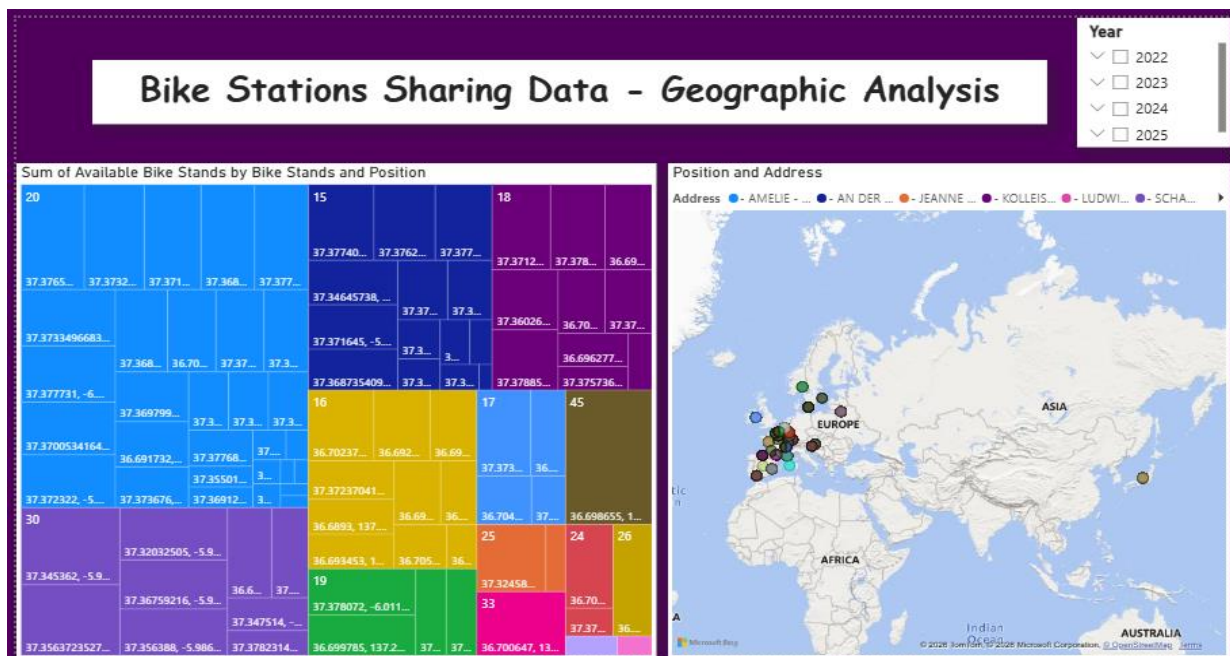
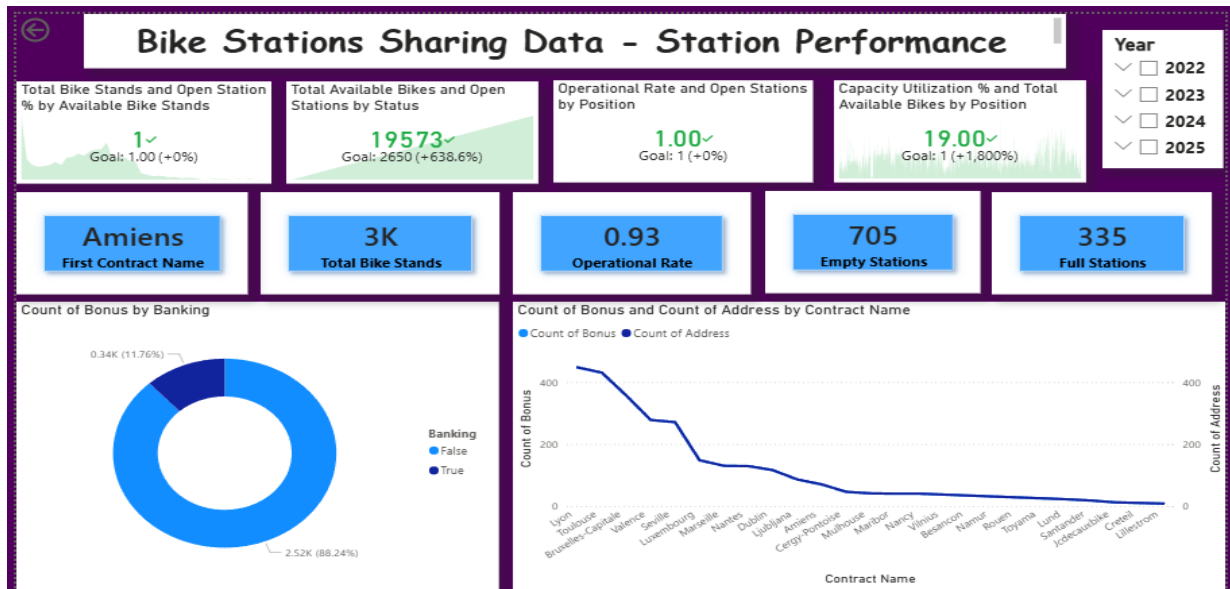
8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **Multiple Visualizations based on problem statement:**
 Developed interactive visualizations using bar charts, line charts, pie/donut charts, KPI cards, maps, treemaps, and tables to present key insights and improve data interpretation.
- This version better reflects the dashboards you've shared, which include:
 1. **KPI Cards**
 2. **Bar Charts**
 3. **Line Charts**
 4. **Pie/Donut Charts**
 5. **Treemap**
 6. **Map Visualization**
 7. **Table/Matrix**
 8. **Slicers (Year filter)**
- Make your Report interactive Layout using **Drill-down, Filters and slicers**
- **Create a consolidated Report /Dashboard.**
 1. **Data Cleaning & Transformation:** Cleaned the dataset by removing duplicates, handling missing values, standardizing data formats, and creating calculated columns to improve data quality and support accurate analysis.
 2. **Filtering & Sorting:** Applied filtering and sorting techniques to organize data, focus on relevant records, and streamline the analytical process.
 3. **Data Modeling:** Established relationships between tables and optimized the data model to support efficient reporting and analysis.
 4. **DAX Measures:** Developed DAX measures and calculated columns to compute KPIs such as Operational Rate, Capacity Utilization, Available Bikes, Empty Stations, and Full Stations.
 5. **Dashboard Development:** Created a consolidated interactive Power BI dashboard integrating KPI cards, charts, maps, tables, and slicers to deliver comprehensive insights into contract performance, station performance, and geographic distribution.

SCREENSHOT OF DASHBOARD /REPORT





9. Insights & Conclusions

• Key Findings: Insights

- Approximately **93% of bike stations are operational**, while only a small percentage are closed, indicating a highly available bike-sharing network.
- The system contains **around 32K bike stands**, with **8.6K bikes currently available**, suggesting that a significant portion of docking capacity is occupied.
- Lyon, Toulouse, and Bruxelles-Capitale** have the highest number of bike stands and stations, making them the largest bike-sharing contracts.
- 705 stations are empty**, requiring immediate bike redistribution to maintain service availability.
- 335 stations are completely full**, indicating a need for bike collection to free docking spaces.
- Most stations do **not support banking/payment facilities**, while only a limited number provide banking services.
- The geographic distribution shows that bike stations are concentrated across **European cities**, with the highest density in France and neighboring countries.

8. Station capacity and bike availability vary considerably between cities, highlighting opportunities for improved resource allocation.

- **Provide the analysis insights:**

Approximately **93% of bike stations are operational**, indicating a reliable bike-sharing network. The system includes around **32,000 bike stands** and **8,600 available bikes**. **Lyon** has the highest number of bike stations, followed by **Toulouse** and **Bruxelles-Capitale**. The network also contains **705 empty stations** and **335 full stations**, while most stations operate without banking facilities. The service is mainly concentrated across **European cities**.

The high number of stations in cities like **Lyon, Toulouse, and Bruxelles-Capitale** is likely due to greater population density and commuter demand. Empty stations are mainly caused by high bike usage or delayed redistribution, while full stations result from bikes being returned faster than they are collected. Differences in bike availability and banking facilities across cities reflect varying demand patterns and operational requirements.

If current trends continue, high-demand cities such as **Lyon** and **Toulouse** are likely to experience more empty stations during peak hours. Frequently full stations may continue to face docking shortages unless bike collection improves. Cities with high utilization may require additional bike stands, while seasonal and daily demand changes will continue to influence bike availability.

Bike redistribution should be prioritized for stations with low availability, while full stations should have bikes collected more frequently. Expanding docking capacity in high-demand cities and using demand forecasting can improve bike availability. Continuous monitoring, automated alerts, and additional banking facilities where needed will help enhance operational efficiency and provide a better user experience.

10. Conclusions:

The **Bike Stations Sharing Data Analysis Dashboard** provides a comprehensive view of station availability, operational performance, and geographic distribution across multiple cities. The analysis reveals that the majority of bike-sharing contracts remain **operational**, indicating a well-maintained network. However, bike availability and station capacity utilization vary significantly across locations, highlighting opportunities for improved resource allocation.

Key findings include:

- Most bike-sharing contracts are **open and operational**, with a high overall operational rate.
- Cities such as **Lyon, Toulouse, and Brussels-Capitale** have the largest number of bike stands and stations, indicating higher service demand.
- Approximately **32K bike stands** exist across all contracts, while only **8.6K bikes are currently available**, suggesting that bike distribution can be optimized.
- Around **705 stations are empty**, requiring urgent bike redistribution, while **335 stations are completely full**, indicating the need for timely bike collection.
- Geographic analysis shows that bike-sharing services are concentrated primarily across **European cities**, with varying levels of station capacity and bike availability.
- Capacity utilization differs between cities, indicating opportunities to improve operational efficiency and balance bike availability across the network.